A PROJECT REPORT

on

AI TEXT DETECTOR (Human vs AI Generated Text Classification)

As a part of

Data Science

[CE0630]

Submitted by

Rishabh Chopda (IU2341230409)

Hardika Nimavt (IU2341230421)

Viren Shah (IU2341230433)

In fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE ENGINEERING

INSTITUTE OF TECHNOLOGY AND ENGINEERING

INDUS UNIVERSITY CAMPUS, RANCHARDA, VIA-THALTEJ

AHMEDABAD-382115, GUJARAT, INDIA,

WEB: www.indusuni.ac.in

1. Abstract

The AI Text Detector project is a machine learning-based system designed to classify textual data as either human-written or AI-generated. With the increasing use of artificial intelligence tools such as ChatGPT, it has become important to identify the origin of textual content.

This system uses Natural Language Processing (NLP) techniques along with machine learning algorithms to analyze patterns in text. The project includes data preprocessing, exploratory data analysis (EDA), feature extraction using TF-IDF, and model training using Logistic Regression and Naive Bayes.

The model is evaluated using performance metrics such as accuracy, confusion matrix, and ROC curve. Additionally, a prediction system is implemented that allows users to input custom text and receive classification results along with a confidence score.

2. Introduction

Artificial Intelligence has transformed content generation. Today, AI can generate essays, articles, and conversations that closely resemble human writing. However, this creates challenges in areas like education, journalism, and online content verification.

The AI Text Detector project focuses on solving this problem by building a classification system that distinguishes between human-written and AI-generated text.

The system uses machine learning algorithms trained on labeled data. By analyzing patterns such as word usage, sentence structure, and frequency of terms, the model learns to differentiate between the two types of text.

This project demonstrates the application of NLP and machine learning in real-world scenarios.

3. Objectives

- To collect and understand textual dataset
- To preprocess text data for better analysis
- To perform exploratory data analysis
- To extract meaningful features using TF-IDF
- To build and train classification models
- To compare different machine learning models
- To evaluate model performance using metrics
- To develop a prediction system for real-time input

4. Technologies Used

With the rapid growth of AI-generated content, it has become difficult to differentiate between human-written and machine-generated text. This creates issues in academic integrity, misinformation, and content authenticity.

The problem is to design a system that can accurately classify text into two categories:

- Human-written
- AI-generated

5. System Architecture & Modules

- Educational institutions (plagiarism detection)
- Content verification systems
- Blogging platforms
- Fake content detection
- AI monitoring tools

5.1 Data Ingestion

The dataset is loaded from a CSV file using Pandas.

It contains text data and corresponding labels (0 = Human, 1 = AI).

Initial inspection is done using functions like head() and info().

5.2 Data Cleaning

- Missing values are checked
- Duplicate records are removed
- Text is converted to lowercase and special characters are removed

This step ensures clean and consistent data.

5.3 Exploratory Data Analysis

EDA is performed to understand the dataset:

- Distribution of AI vs Human text
- Text length and word count analysis

Visualizations like histograms and bar charts are used.

5.4 Preprocessing Pipeline

- Features (text) and labels are separated
- Train-test split (80-20) is applied
- TF-IDF is used to convert text into numerical form

5.5 Model Training

Machine learning models used:

- Logistic Regression
- Naive Bayes

These models learn patterns from the training data.

5.6 Evaluation & Feature Importance

Model performance is evaluated using:

- Accuracy
- Confusion Matrix
- Classification Report

Important features (words) are analyzed to understand model decisions.

5.7 Real-Time Prediction Interface

The system takes user input text and:

- Processes the text
- Applies the trained model

Output:

- Human-written or AI-generated text
- Confidence score

6. Key Concepts Used

Technology	Description
Python	Core programming language
Pandas	Data manipulation and cleaning

NumPy	Numerical computations
Matplotlib	Graph plotting
Seaborn	Statistical visualization
Scikit-learn	Machine learning models
NLP	Text processing techniques

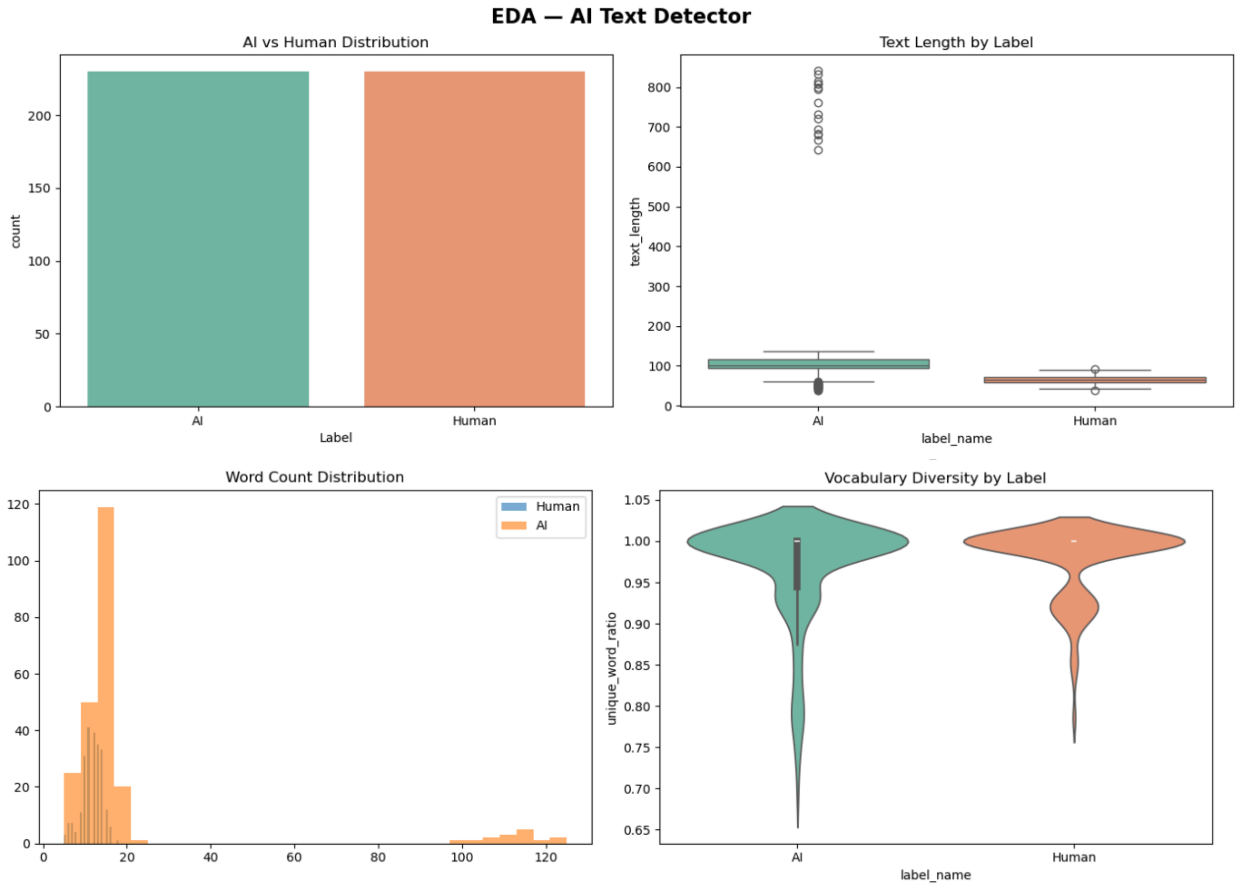
7. Screenshots

```
Dataset shape: (460, 2)
Label distribution:
label
1    230
0    230
Name: count, dtype: int64
```

	text	label
0	Exercise plays a crucial role in supporting me...	1
1	Renewable energy helps fight climate change by...	1
2	A futuristic smart city is a vibrant, intercon...	1
3	Healthy eating habits are especially important...	1
4	Machine learning is transforming healthcare by...	1

```
Feature engineering done
```

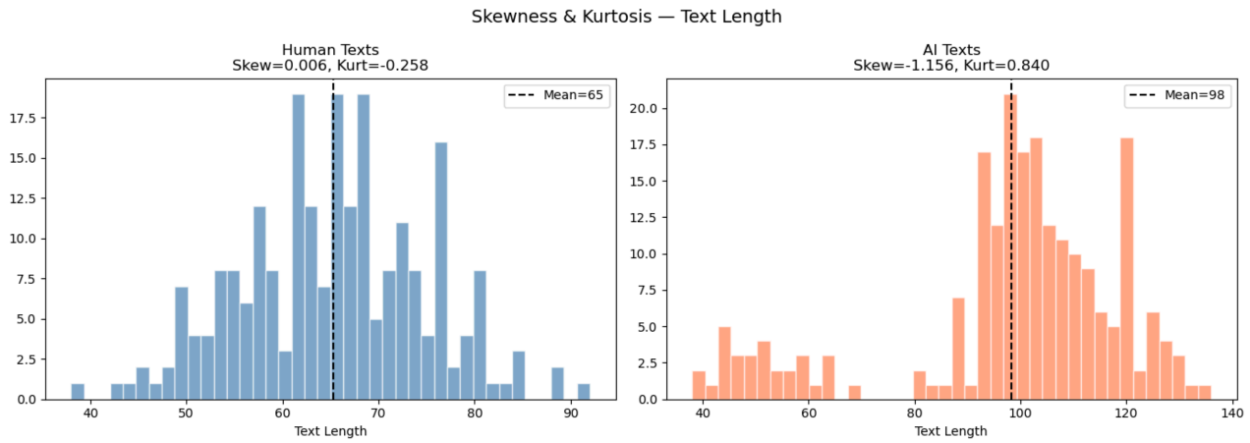
	text_length	word_count	avg_word_length	unique_word_ratio
count	460.000000	460.000000	460.000000	460.000000
mean	103.073913	15.560870	5.736074	0.972748
std	121.980743	18.019675	1.270744	0.054447
min	38.000000	5.000000	3.250000	0.695652
25%	63.000000	11.000000	4.685897	0.944444
50%	76.000000	13.000000	5.732057	1.000000
75%	102.000000	14.000000	6.723214	1.000000
max	841.000000	125.000000	10.111111	1.000000



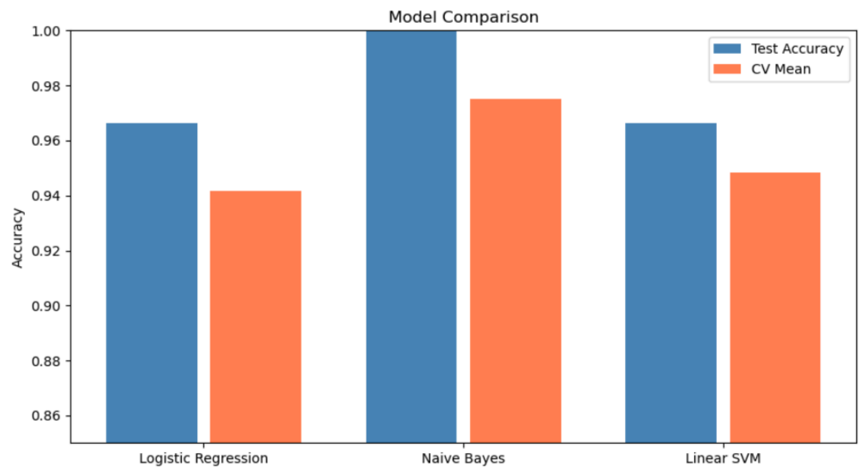

```
============================================================
 STATISTICAL ANALYSIS OF NUMERICAL FEATURES
============================================================

text_length
   Mean     : 103.0739
   Std      : 121.9807
   Skewness : 5.0420   → Right-Skewed (positive tail)
   Kurtosis : 24.7371   → Leptokurtic (heavy tails / sharp peak)

word_count
   Mean     : 15.5609
   Std      : 18.0197
   Skewness : 5.1029   → Right-Skewed (positive tail)
   Kurtosis : 24.9211   → Leptokurtic (heavy tails / sharp peak)

avg_word_length
   Mean     : 5.7361
   Std      : 1.2707
   Skewness : 0.1933   → Approximately Symmetric
   Kurtosis : -0.6473   → Mesokurtic (normal-like)

unique_word_ratio
   Mean     : 0.9727
   Std      : 0.0544
   Skewness : -2.3026   → Left-Skewed (negative tail)
   Kurtosis : 5.3473   → Leptokurtic (heavy tails / sharp peak)
```

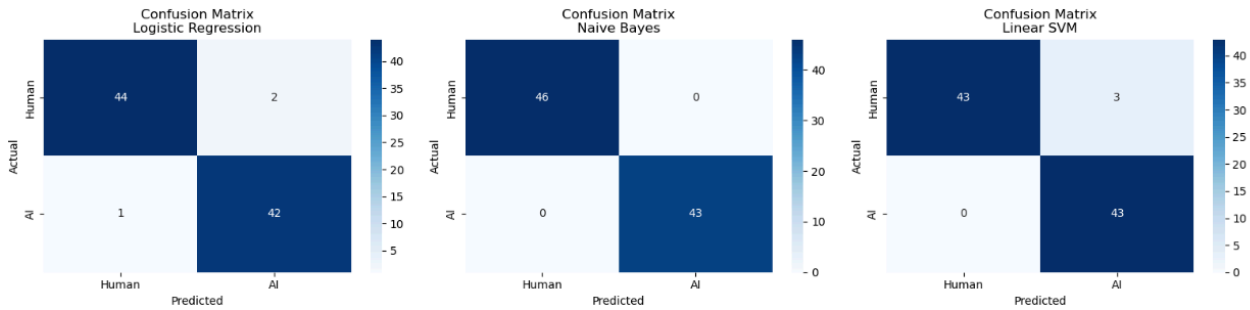
```
============================================================
 Z-SCORE OUTLIER DETECTION  (|z| > 3 = outlier)
============================================================
text_length               : 15 outliers (3.26%)
word_count                : 15 outliers (3.26%)
avg_word_length           : 1 outliers (0.22%)
unique_word_ratio         : 16 outliers (3.48%)

Dataset after removing text_length outliers: 445 rows
```

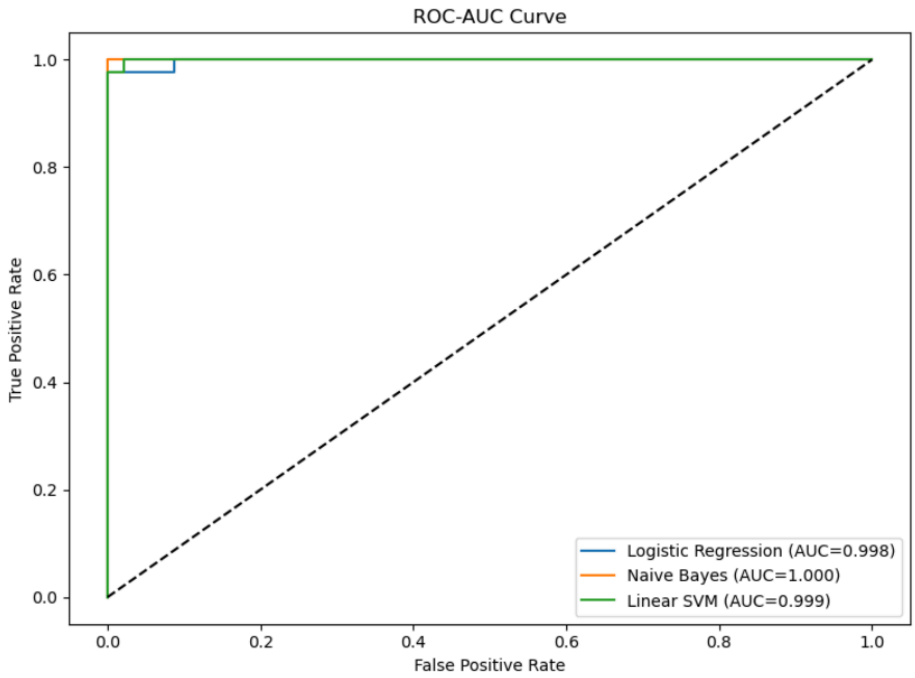
```
=======================================================
 TWO-SAMPLE Z-TEST: AI vs Human — text_length
=======================================================
  AI Mean     : 98.22
  Human Mean  : 65.30
  Z-Statistic: 20.8601
  P-Value     : 0.000000

 Result: Statistically significant difference (p < 0.05)
AI and Human texts differ in length significantly.

=======================================================
 TWO-SAMPLE Z-TEST: AI vs Human — word_count
=======================================================
  Z-Statistic: 5.3036  |  P-Value: 0.000000
Significant difference in word count between AI and Human texts.
```

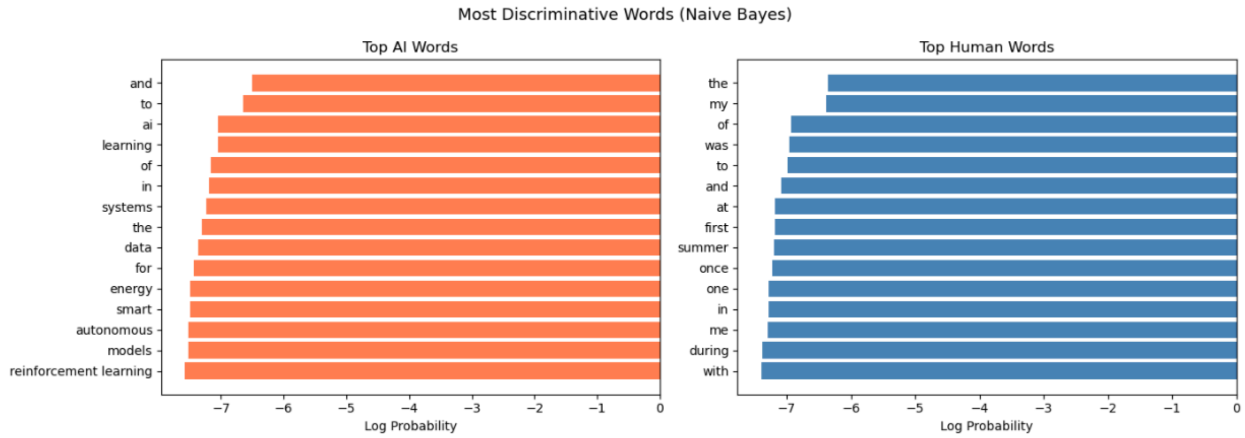


```
--- Logistic Regression ---
  Test Accuracy : 0.9663
  CV (5-fold)   : 0.9416 ± 0.0892
              precision    recall  f1-score   support

       Human       0.98      0.96      0.97        46
          AI       0.95      0.98      0.97        43

    accuracy                           0.97        89
   macro avg       0.97      0.97      0.97        89
weighted avg       0.97      0.97      0.97        89

--- Naive Bayes ---
  Test Accuracy : 1.0000
  CV (5-fold)   : 0.9753 ± 0.0385
              precision    recall  f1-score   support

       Human       1.00      1.00      1.00        46
          AI       1.00      1.00      1.00        43

    accuracy                           1.00        89
   macro avg       1.00      1.00      1.00        89
weighted avg       1.00      1.00      1.00        89

--- Linear SVM ---
  Test Accuracy : 0.9663
  CV (5-fold)   : 0.9483 ± 0.0925
              precision    recall  f1-score   support

       Human       1.00      0.93      0.97        46
          AI       0.93      1.00      0.97        43

    accuracy                           0.97        89
   macro avg       0.97      0.97      0.97        89
weighted avg       0.97      0.97      0.97        89
```










```
Enter text to detect:
 You are the beautiful girl I had seen till now.
===== RESULT =====
Label      : Human Written
AI %       : 0.0%
Confidence : 57.22%
```

```
(np.int64(0), 0.0, np.float64(0.5721907706588527))
```

```
============================================================
  PROJECT SUMMARY — AI Text Detector
============================================================
Dataset size     : 445 samples (balanced)
Features used    : TF-IDF (5000 features, unigrams+bigrams)

MODEL RESULTS:
  Logistic Regression       : Test=0.9663, CV=0.9416±0.0892
  Naive Bayes               : Test=1.0000, CV=0.9753±0.0385
  Linear SVM                : Test=0.9663, CV=0.9483±0.0925

STATISTICAL HIGHLIGHTS:
  • Skewness & Kurtosis computed for all numerical features
  • Z-Score based outlier removal applied
  • Two-sample Z-Test confirms AI/Human text length difference
```

7. Achievements, Limitations & Future Scope

7.1 Achievements

- Successfully developed an end-to-end machine learning pipeline for text classification
- Implemented data preprocessing and feature extraction using TF-IDF
- Trained multiple models such as Logistic Regression and Naive Bayes
- Achieved good accuracy in distinguishing AI-generated and human-written text
- Performed exploratory data analysis using graphs and statistical methods
- Built a real-time prediction system for user input text

7.2 Limitations

- Limited dataset size may affect model accuracy and generalization
- Model may not perform well on mixed or edited text
- No advanced deep learning models used
- Accuracy depends heavily on quality of training data
- Limited feature engineering applied

7.3 Future Scope

- Implement advanced models like BERT, LSTM, or Transformer-based models
- Use larger and more diverse datasets
- Develop a web-based or mobile application
- Improve accuracy using hyperparameter tuning
- Integrate real-time API for practical applications

8. Conclusion

The AI Text Detector project demonstrates how machine learning and NLP can be used to solve real-world problems. The system effectively classifies text and provides accurate predictions.

With further improvements, this system can be used in educational and professional environments for content verification.